International Institute of Information Technology Bangalore

PAVITHRA KUNE
COMET FWC 69

GATE Mathematics Questions (1986–1988)

1986/1

Let d be any positive integer not equal to 2, 5, or 13.
Show that one can find distinct a, b in the set

$$\{2, 5, 13, d\}$$

such that

$$ab - 1$$

is not a perfect square.

1986/2

A triangle $A_1A_2A_3$ and a point P_0 are given in the plane.
Define

$$A_s = A_{s-3}$$

for all $s \geq 4$.
Construct points $P_1, P_2, \ldots$ such that P_{k+1} is the image of P_k under a rotation with center A_{k+1} through angle 120° clockwise.
Prove that if

$$P_{1986} = P_0$$

then the triangle $A_1A_2A_3$ is equilateral.

1986/3

To each vertex of a regular pentagon an integer is assigned so that the sum of all five numbers is positive.

If three consecutive vertices are assigned x, y, z with $y < 0$, then the operation

$$(x, y, z) \mapsto (x + y, -y, z + y)$$

is allowed.

Determine whether this procedure necessarily ends after finitely many steps.

1986/4

Let A, B be adjacent vertices of a regular n-gon ($n \geq 5$) centered at O.

A triangle XYZ, congruent to OAB, moves so that Y, Z trace the boundary of the polygon while X remains inside.

Find the locus of X.

1986/5

Find all functions

$$f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$$

such that

$$f(xf(y))f(y) = f(x + y)$$

and

$$f(2) = 0$$

with

$$f(x) > 0$$

for

$$0 \leq x < 2$$

1986/6

Given a finite set of integer-coordinate points in the plane, determine whether it is always possible to color them red and white so that for any line parallel to an axis, the difference between the number of red and white points is at most 1.

1987/1

Let $P_n(k)$ be the number of permutations of

$$\{1, \ldots, n\}$$

with exactly k fixed points.
Prove that

$$\sum_{k=0}^{n} kP_n(k) = n!$$

1987/2

In an acute triangle ABC, the bisector of $\angle A$ meets BC at L and the circumcircle again at N.
From L, draw perpendiculars to AB and AC with feet K and M respectively.
Prove that quadrilateral $AKNM$ and triangle ABC have equal areas.

1987/3

Let

$$x_1, \ldots, x_n$$

be real numbers satisfying

$$x_1^2 + \cdots + x_n^2 = 1$$

Prove that for every integer

$$k \geq 2$$

there exist integers

$$a_1, \ldots, a_n$$

not all zero, with

$$|a_i| < k - 1$$

and

$$|a_1 x_1 + \cdots + a_n x_n| < \frac{(k-1)\sqrt{n}}{kn - 1}$$

1987/4

Prove that no function

$$f : \mathbb{Z}_{\geq 0} \to \mathbb{Z}_{\geq 0}$$

satisfies

$$f(f(n)) = n + 1987$$

1987/5

Prove that for any integer

$$n \geq 3$$

there exists a set of n points in the plane such that distances are irrational, yet every triple forms a triangle of rational area.

1987/6

Let

$$n \geq 2$$

Prove that if

$$k^2 + k + n$$

is prime for all integers

$$0 < k \leq \frac{\sqrt{n}}{3}$$

then it is prime for all

$$0 \leq k \leq n - 2$$

1988/1

Two concentric circles have radii $R > r$.

Point P lies on the smaller circle and point B lies on the larger circle.

Line BP meets the larger circle again at C.

The perpendicular at P meets the smaller circle again at A.

Find:

- values of

$$BC^2 + CA^2 + AB^2$$

- locus of midpoint of BC

1988/2

Let

$$A_1, \ldots, A_{2n+1}$$

be subsets of a set B such that:

- each has $2n$ elements
- each pair intersects in exactly one element
- every element of B belongs to at least two subsets

Determine for which n one can assign 0 or 1 to elements so that each A_i has exactly n zeros.

1988/3

Define f on positive integers by

$$f(1) = 1$$

$$f(3) = 3$$

$$f(2n) = f(n)$$

$$f(4n + 1) = 2f(2n + 1) - f(n)$$

$$f(4n+3) = 3f(2n+1) - 2f(n)$$

Determine the number of

$$n \leq 1988$$

such that

$$f(n) = n$$

1988/4

Show that the set of real numbers x satisfying

$$\sum_{k=1}^{70} \frac{x-k}{k^5} \geq 4$$

is a union of disjoint intervals whose total length is 1988.

1988/5

In right triangle ABC where

$$\angle A = 90^\circ$$

let D be the foot of altitude from A.

The line joining incenters of triangles ABD and ACD meets AB and AC at K and L.

If S and T are areas of triangles ABC and AKL respectively, show that

$$S \geq 2T$$

1988/6

Let a, b be positive integers such that

$$ab + 1 \mid a^2 + b^2$$

Show that

$$\frac{a^2+b^2}{ab+1}$$

is a perfect square.